

Third Terminal Examination - 2079

Class : XI

Subject: Biology (Code: 2011)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt All Questions

PART : I [BOTANY]

Group 'A' [BOTANY]

Rewrite the best alternative of each question in your answer sheet.

[6 × 1 = 6]

1. The following statement is wrong for a monocot plant
 - a. It has parallel veined leaves
 - b. It has adventitious root system
 - c. It has pentamerous flower
 - d. It has single cotyledon in its seed
2. 80S ribosomes has two sub-units, which are
 - a. 50S+30S
 - b. 60S+40S
 - c. 60S+20S
 - d. 60S+50S
3. The outermost covering of vacuole is called
 - a. Tonoplast
 - b. Cell membrane
 - c. Plasma membrane
 - d. Leucoplast
4. The study of individual population in ecology is called
 - a. Autecology
 - b. Synecology
 - c. Ecosystem
 - d. Phenology
5. The following is also known as "cytoskeleton"
 - a. Golgi bodies
 - b. Endoplasmic reticulum
 - c. Mitochondria
 - d. Cell wall
6. When sepals & petals are indistinguishable, they are collectively called
 - a. Tepals
 - b. Petals
 - c. Calyx
 - d. Perianth

Group 'B' [BOTANY]

Give short answer to the following questions.

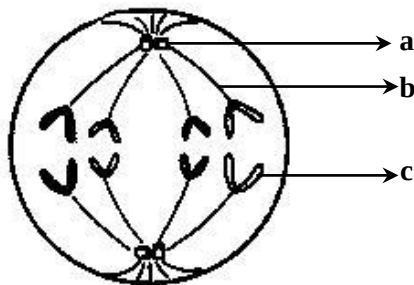
[4 × 4 = 16]

1. Provide scientific term for each. [1 × 4]
 - a. Spindle shaped taproot that stores food.
 - b. Arrangement of leaves on stem.
 - c. Horizontally growing underground stem with nodes and internodes and covered by scaly leaves.
 - d. Stamens are arranging in two bundle of 1+(9) is called
2. Define autecology. Mention the roles of temperature in an ecosystem. [1+3 = 4]
3. Define binomial system. State the rules of binomial nomenclature. [1+3 = 4]

OR

Why is mitochondrion called power house of a cell? Enlist the functions of golgi bodies. [1+3]

4. Answer the following questions based on figure given. [1×4 = 4]
 - a. What stage of mitosis is the figure about?
 - b. Name the labelled parts a, b & c.



Group 'C' [BOTANY]

Give long answer to the following questions.

[2 × 8 = 16]

5. Describe the family Brassicaceae in semi-technical terms with floral formula and floral diagram. State the scientific names of any two economic plants of the family. [3+1+3+1=8]

OR

Why are bryophytes called amphibian plants? Describe the life cycle of *Marchantia*. [1+7 = 8]

6. Define ecosystem. What are abiotic factors? Explain the roles of light, temperature and wind in an ecosystem. [1+1+2+2+2 = 8]

PART : II [ZOOLOGY]

Group 'A' [ZOOLOGY]

Rewrite the best alternative of each question in your answer sheet.

[5 × 1 = 5]

1. The inheritance of characters from parents to offspring is called
 - a. Heredity
 - b. Variation
 - c. Cytology
 - d. Evolution
2. Life was first originated in
 - a. Sea water
 - b. Air
 - c. Land
 - d. All of these
3. *Archaeopteryx* is connecting link between
 - a. Reptile & Bird
 - b. Birds & Amphibia
 - c. Annelida & Arthropoda
 - d. Reptile & Mammal
4. Which of the followings is the infective stage of malarial parasite?
 - a. Cryptozoite
 - b. Merozoite
 - c. Sporozoite
 - d. Trophozoite
5. Forest of nephridia is in
 - a. Clitellar region
 - b. Pharyngeal region
 - c. Typlosolar region
 - d. Intestinal region

Group 'B' [ZOOLOGY]

Give short answers to the following questions.

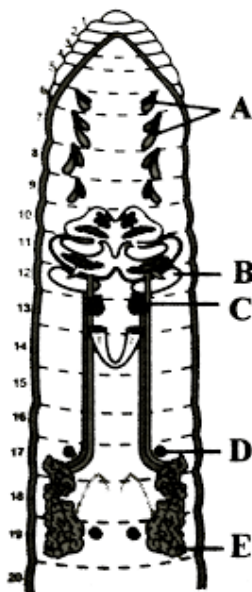
[4 × 4 = 16]

1. Briefly discuss the scopes of biology in this modern day. [4]
2. Why are most mosquitoes now found to be DDT resistant although DDT was supposed to be highly effective insecticide in the past? [2+2]

OR

Describe the erythrocytic schizogony of *Plasmodium vivax* with well labeled diagram. [2+2]

3. The following is the diagram of reproductive system of earthworm. Answer the following:



- a. Mention A, C, D, E. [4]
4. Differentiate between hibernation & aestivation. [4]

Group 'C' [ZOOLOGY]

Give long answers to the following questions. [2 × 8 = 16]

5. What is conjugation? Describe conjugation in *Paramecium caudatum* with diagrams. Mention any two significances of conjugation. [1+4+2+1]

OR

What is endonephric nephridia? Describe the structure of Septal nephridia with labelled diagram. [1+4+3]

6. Name the theory of evolution proposed by Darwin. Enumerate the key concepts in the evolution theory of Darwin. [1+7]

The End